
Q1. Display all film titles and their rental rates.

Query:

```
SELECT title, rental_rate
```

```
FROM film;
```

Explanation:

Simple selection of two columns from the film table.

Q2. List all films released in the year 2006.

Query:

```
SELECT title, release_year
```

```
FROM film
```

```
WHERE release_year = 2006;
```

Explanation:

Filters records using a WHERE condition.

Q3. Find all customers who are active.

Query:

```
SELECT customer_id, first_name, last_name
```

```
FROM customer
```

```
WHERE active = 1;
```

Explanation:

Displays only customers whose active status equals 1.

Q4. List the top 10 longest films.

Query:

```
SELECT title, length
```

```
FROM film
```

```
ORDER BY length DESC
```

```
LIMIT 10;
```

Explanation:

Orders films by length (descending) and shows the first 10.

Q5. Find the number of films for each rating.**Query:**

```
SELECT rating, COUNT(*) AS total_films
```

```
FROM film
```

```
GROUP BY rating;
```

Explanation:

Groups films by rating (like G, PG, PG-13) and counts them.

Q6. Find the total number of active and inactive customers.**Query:**

```
SELECT active, COUNT(*) AS total_customers
```

```
FROM customer
```

```
GROUP BY active;
```

Explanation:

Groups by the active column to count how many are active vs inactive.

Q7. Find all films whose title starts with 'A'.**Query:**

```
SELECT title
```

```
FROM film
```

```
WHERE title LIKE 'A%';
```

Explanation:

Uses the LIKE operator to match titles beginning with "A".

Q8. Find all films whose rental rate is between 1.00 and 3.00.**Query:**

```
SELECT title, rental_rate  
FROM film  
WHERE rental_rate BETWEEN 1.00 AND 3.00;
```

Explanation:

Uses BETWEEN to filter rental_rate within a range.

Q9. Find all customers whose last name ends with 'S'.**Query:**

```
SELECT first_name, last_name  
FROM customer  
WHERE last_name LIKE '%S';
```

Explanation:

Uses LIKE with %S to match any last name ending with "S".

Q10. Find film categories that have more than 200 films (use HAVING).

Note: If you're only allowed one table, we'll use film table grouping by rating.
So we'll slightly adjust:

Query:

```
SELECT rating, COUNT(*) AS total_films  
FROM film  
GROUP BY rating  
HAVING COUNT(*) > 200;
```

Explanation:

Uses HAVING to filter grouped results after aggregation (ratings with >200 films).